Are Repeated Novel Geometries Addressable Without Reasoning? A Controlled Study of Direct vs. Chain-of-Thought Prompting on Verbal Grid Addressing

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Abstract

When a large language model is shown a novel 2D shape rendered as a text grid, can it verbally address that shape—name what is at a cell, locate a token, count occurrences, find the densest row—without producing an explicit reasoning trace? We design GRIDADDRESS, a controlled $2\times2\times5$ factorial study over two models (GPT-4.1-MINI and CLAUDE-HAIKU-4.5), two prompting regimes (DIRECT and COT), and five levels of in-context repetition ($K{=}0, 1, 2, 4, 8$ same-grid demonstrations), evaluated on 80 procedurally-generated 7×7 ASCII grids and five question types with mechanical ground truth. We find that DIRECT prompting plateaus at 79–93% accuracy even with $K{=}8$ demos, while COT lifts both models to 99–100% from $K{=}0$. A per-question-type breakdown shows the gap is driven almost entirely by two scanning queries (COUNT and ROWMAX): random-access queries (POINT, LOOKUP, NEIGHBOR) are at near-ceiling under DIRECT prompting. Paired McNemar tests confirm the DIRECT–COT gap at every (model, K) cell ($p{\leq}0.031$, nine of ten significant under Bonferroni at $\alpha{=}0.005$). Same-grid repetition matters under DIRECT ($0.65 \rightarrow 0.93$ for CLAUDE-HAIKU-4.5, $p{<}10^{-6}$), but cannot close the aggregation gap that COT closes for free. The picture refines the "curse of CoT" story for in-context learning: reasoning helps strictly when the addressing query requires a scan, and is the cheapest reliability lever for any LLM workflow grounded in small text grids.

1 Introduction

Many practical uses of large language models embed the model in a small, discrete environment that is best described as a text grid: a screenshot serialised as ASCII, a board game state, an ARC-style puzzle, a tabular form. In all of these settings the model must *address* the grid verbally—report the token at a cell, locate a feature, count occurrences, identify the densest row. Whether this kind of addressing works *without* explicit step-by-step reasoning has direct consequences for inference cost, latency, and the design of agentic systems that ground decisions in serialised state.

A growing body of work argues that chain-of-thought (CoT) prompting often hurts on in-context pattern-induction tasks [Zheng et al., 2025, Sprague et al., 2025], and that text-only LLMs can pick up 2D-geometric structure from a few in-context demonstrations [Liu et al., 2025]. A parallel line of work shows that vision-language models fail on novel polygons unless prompts label the geometry explicitly [Rudman et al., 2025, Wang et al., 2025]. And surface-level repetition in the prompt is known to push next-token predictions toward self-reinforced patterns [Yan et al., 2024]. **What none of these threads has done is hold the geometry fixed, vary the amount of in-context repetition about that same geometry, and ask whether direct prompting can substitute for reasoning on verbal addressing of a *novel* shape.** That intersection is the gap we close.

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Our approach. We design GRIDADDRESS, a controlled $2{\times}2{\times}5$ factorial study (figure 1). We procedurally generate 80 novel $7{\times}7$ ASCII grids, each containing one irregular shape grown by a random walk over an alphabet of four non-empty tokens. For every grid we issue one held-out target addressing question drawn from five mechanically-checkable question types (POINT, LOOKUP, COUNT, ROWMAX, NEIGHBOR), padded by $K \in \{0, 1, 2, 4, 8\}$ same-grid demonstration (Q, A) pairs. We sweep two models (GPT-4.1-MINI, CLAUDE-HAIKU-4.5) and two prompting regimes (DIRECT, COT) at each K, for 1,600 total queries. The design lets us cleanly factor out the effect of reasoning, repetition, and query type with within-item paired statistics.

Headline result. DIRECT prompting plateaus at 79–93% accuracy even at $K{=}8$, while COT lifts both models to 99–100% from $K{=}0$ onward. A per-query-type breakdown reveals the cause: random-access queries (POINT, LOOKUP, NEIGHBOR) succeed at $\geq 95\%$ accuracy under DIRECT, while COUNT and ROWMAX—queries that require an iterative scan of the grid—fail 12–68% of the time. Paired McNemar tests show the DIRECT–COT gap is significant at every (model, K) cell ($p \leq 0.031$), and nine of ten survive a Bonferroni correction over the ten paired tests. Repeating the same grid helps DIRECT prompting (CLAUDE-HAIKU-4.5: $0.65 \rightarrow 0.93$, $p < 10^{-6}$) but *cannot* close the aggregation gap that COT closes for free.

Why this matters. Our results refine the "curse of CoT" story for in-context learning: the curse holds for rule induction but inverts for verbal addressing of novel geometries, with the inversion driven by exactly the scan-and-accumulate queries that Sprague et al. [2025] identify as the regime where CoT actually buys accuracy. For practitioners building LLM agents on top of grid-shaped state, enabling CoT is a strictly cheaper reliability lever than padding the context with same-grid demonstrations.

Contributions.

- We propose GRIDADDRESS, a procedurally-generated benchmark of verbal addressing on novel 2D ASCII shapes that factors reasoning against in-context repetition with within-item paired statistics.
- We show that two production LLMs reach 99–100% accuracy under COT from $K{=}0$, while DIRECT prompting plateaus at 79–93% even at $K{=}8$, and identify scan-based aggregation queries (COUNT, ROWMAX) as the sole source of the gap.
- We complement the result with a MINIARC sanity check showing GPT-4.1-MINI exhibits the opposite DIRECT $>$COT ordering on rule induction, reconciling our finding with the curse-of-CoT literature: the gap depends on whether the query is addressing or induction.
- We release code, prompts, raw model outputs, and analysis scripts at total reproducibility cost of about \$1.50.

Section 2 surveys CoT skepticism, spatial reasoning in LLMs, and in-context repetition effects. Section 3 describes GRIDADDRESS and the statistical analysis plan. Section 4 presents the main results. Section 5 interprets them and discusses limitations. Section 6 concludes.

2 Related Work

Our work sits at the intersection of three recent threads: scepticism about chain-of-thought on in-context tasks, the limits of LLM spatial reasoning, and surface-repetition effects in prompts.

When does chain-of-thought help? Zheng et al. [2025] test 16 LLMs across nine pattern-induction benchmarks and report a *Curse of CoT*: direct prompting beats CoT by 20% on average and 42% on symbolic-grid tasks, with the gap widening as more demonstrations are added. The proposed mechanism is that CoT injects noise via flawed rationales while increasing the contextual distance between demonstrations and the answer, disrupting the implicit ICL pathway. Sprague et al. [2025] provide a complementary meta-analysis: across 20 datasets and 14 LLMs, CoT substantially helps only on math and symbolic-execution tasks, with 95% of CoT's gain on MMLU concentrated in questions containing the "=" symbol. Unlike both lines, which treat pattern induction holistically and conclude that CoT hurts, we hold the geometry fixed and ask about *addressing* a known grid rather than inducing a rule; the per-query-type analysis we report (section 4) shows that the two pictures are consistent once one separates scan-and-aggregate queries from random-access queries.

LLM spatial reasoning and shape understanding. Rudman et al. [2025] show that multimodal LLMs fall to near-chance accuracy on novel polygons and that perception, not language, is the bottleneck: their VC-CoT intervention—visually labelling vertices with letters and asking the model to enumerate them—lifts GPT-4o from 7% to 93% on irregular-polygon side counts. Wang et al. [2025] introduce Primal Visual Description, a vertex/colour JSON schema, and show that a text-only LLM consuming PVD beats GPT-4V on shape comparison and maze tasks. Wu et al. [2024] propose Visualization-of-Thought (VoT) for text-grid navigation, re-rendering the grid at each step. Bai et al. [2025] stress-test Claude 3.7 Sonnet (16k thinking tokens vs. no-thinking), GPT-4o, and GPT-4.1 on five ASCII-grid spatial tasks and find accuracy drops up to 84% as grid size grows. We use the same family of text-grid inputs, but our axis is the *type of question* asked about a fixed grid rather than *grid size* or *representation*; we show that under CoT the question-type axis collapses to ceiling for both tested models at 7×7.

Repetition and in-context learning. Yan et al. [2024] show that ten repetitions of any token sequence push its predicted probability to $\approx$1.0, evidence of a strong self-reinforcement effect that is purely surface-statistical. Liu et al. [2025] show that LLMs pick up 2D-geometric structure from a few in-context (point, value) pairs, and Zheng et al. [2024] provide an axiomatic decomposition of ICL into memorisation and reasoning components. Curse-of-CoT treats demonstration count as a confound rather than a hypothesis variable. We deliberately make K an axis (cf. section 3), so that the joint effect of reasoning and repetition can be read off the $2\times2\times5$ design directly. To our knowledge no prior work crosses verbal addressing of novel 2D shapes with both axes simultaneously.

Novelty in evaluation. ARC-AGI [Chollet, 2019] and its derivatives (MiniARC, 1D-ARC) operationalise novelty through hand-crafted and synthetic rule families. We complement those benchmarks by generating *shapes* rather than *rules*, with a SHA1 hash on rendered grids ensuring no two grids in the experiment are identical, and by asking addressing questions rather than transformations.

3 Methodology

3.1 Task: GRIDADDRESS

GRIDADDRESS consists of procedurally-generated 7×7 ASCII grids, each containing one connected irregular shape grown by a uniform random walk (size 6–12 cells). Non-empty cells are coloured uniformly at random from the alphabet $\{R, B, G, Y\}$; "`.`" marks empty. Each grid is identified by a SHA1 hash of its rendered string, and we verify that all 80 grids in the experiment are pairwise distinct. The grid is rendered into the prompt with single spaces between cells to prevent BPE merges from collapsing cell symbols into bigrams.

For each grid we generate addressing questions from five families, all with mechanical ground truth:

- POINT: "What token is at row r, column c?" $\rightarrow$ single character.
- LOOKUP: "At which (row, col) is the first `X` in row-major order?" $\rightarrow$ coordinate.
- COUNT: "How many cells contain the token `X`?" $\rightarrow$ integer.
- ROWMAX: "Which row has the most non-empty cells (smallest index on ties)?" $\rightarrow$ integer.
- NEIGHBOR: "What token is directly to the right of (r, c)?" $\rightarrow$ character.

POINT, LOOKUP, and NEIGHBOR are random-access queries that the model can in principle resolve by indexing the prompt directly; COUNT and ROWMAX require an iterative scan with an accumulator.

3.2 Prompt structure and reasoning conditions

Each prompt consists of the rendered grid, K demonstration (Q, A) pairs about the same grid (drawn round-robin across the five query types, excluding the target question), and the target question. We compare two reasoning conditions:

- DIRECT: the system prompt instructs the model to emit only the answer (max 24 tokens); demos are formatted as "`A:` ⟨`answer`⟩".
- COT: the system prompt asks for brief reasoning ending in "`Final answer:` ⟨`answer`⟩" (max 256 tokens); demos use the same suffix so the response format is consistent across conditions.

A tolerant regex parses the final answer for both conditions, accepting the *last* coordinate, integer, or character that appears in the response. The parser was sanity-checked on 200 sampled outputs.

3.3 Experimental design

We run a fully crossed $2 \times 2 \times 5$ factorial:

- **Model:** GPT-4.1-MINI (OpenAI), CLAUDE-HAIKU-4.5 (Anthropic via OpenRouter).
- **Reasoning:** DIRECT, COT.
- **Repetition:** $K \in \{0, 1, 2, 4, 8\}$.

Both models are called with temperature 0.0. The eight-demo pool for each grid is sampled once and `demos[:K]` is used at each level, so comparisons across K are within-grid paired. The target question varies across grids round-robin across the five query types. Total queries: $80 \times 5 \times 2 \times 2 = 1600$. Responses are cached on disk keyed by (provider, model, messages, params) so reruns are free.

3.4 Evaluation metrics and statistical analysis

Per cell of the design we report exact-match accuracy with 95% bootstrap confidence intervals (1,000 resamples). For the DIRECT–COT contrast at each (model, K) cell we use a paired McNemar exact test on the within-item binary outcomes; for the K=0 vs. K=8 contrast within reasoning condition we use the same test. Spearman ρ on K vs. correctness summarises the overall trend. We apply Bonferroni correction at family-level α=0.05 for the ten paired DIRECT–COT tests, giving a per-test threshold of α_i=0.005. We additionally report a *cost-normalised* view (accuracy per output token) to ensure the COT accuracy advantage is not bought entirely with tokens.

3.5 Secondary sanity check: MINIARC

To validate our basic setup against a known result we run a small sanity check on 25 MINIARC tasks [CoT-ICL-Eval Contributors, 2024] with the same two models and prompting regimes (no repetition manipulation; the training pairs are intrinsic to each task). The MINIARC setup is a diagnostic for rule-induction, where curse-of-CoT predicts DIRECT $\geq$ COT, and lets us confirm or refute that prediction with the same code path.

3.6 Implementation

Code is implemented in Python with deterministic seeding (grid_seed = 10,000 + i for the i-th grid). Grids are generated by `src/geometry.py`; prompts and the tolerant parser live in `src/prompting.py`; the cached client in `src/llm_client.py`; the sweep driver in `src/run_experiment.py`; analysis (tables, bootstraps, McNemar, plots) in `src/analyze.py`. Total API cost for the published numbers was approximately $1.50.

4 Results

4.1 Headline accuracy

Table 1 reports per-cell exact-match accuracy with 95% bootstrap confidence intervals for the full $2 \times 2 \times 5$ design. The pattern is clean: COT lifts both models to 99–100% accuracy from K=0, while DIRECT prompting plateaus at 79–93% even after eight same-grid demonstrations. Figure 1 plots the same numbers as a function of K.

4.2 Paired DIRECT vs. COT tests

Table 2 reports McNemar exact tests on the paired within-item binary outcomes at every (model, K) cell. COT beats DIRECT at every one of the ten cells; nine of ten survive the Bonferroni-corrected threshold α_i=0.005. The COT-only discordant column is uniformly larger than the DIRECT-only column at every cell.

Table 1: Per-cell exact-match accuracy on GRIDADDRESS (n=80 per cell, 95% bootstrap CIs). COT is at or near ceiling across both models and every K; DIRECT plateaus well below ceiling. Best per row in **bold**.

Model	Reasoning	K=0	K=1	K=2	K=4	K=8
GPT-4.1-MINI	DIRECT	0.788 [0.70, 0.88]	0.812 [0.71, 0.90]	0.825 [0.74, 0.90]	0.850 [0.76, 0.93]	0.875 [0.80, 0.94]
GPT-4.1-MINI	COT	**1.000** [1.00, 1.00]	0.988 [0.96, 1.00]	0.988 [0.96, 1.00]	**1.000** [1.00, 1.00]	**1.000** [1.00, 1.00]
CLAUDE-HAIKU-4.5	DIRECT	0.650 [0.55, 0.76]	0.700 [0.60, 0.80]	0.762 [0.68, 0.85]	0.875 [0.80, 0.94]	0.925 [0.88, 0.98]
CLAUDE-HAIKU-4.5	COT	**0.938** [0.88, 0.99]	**1.000** [1.00, 1.00]	**1.000** [1.00, 1.00]	**1.000** [1.00, 1.00]	**1.000** [1.00, 1.00]

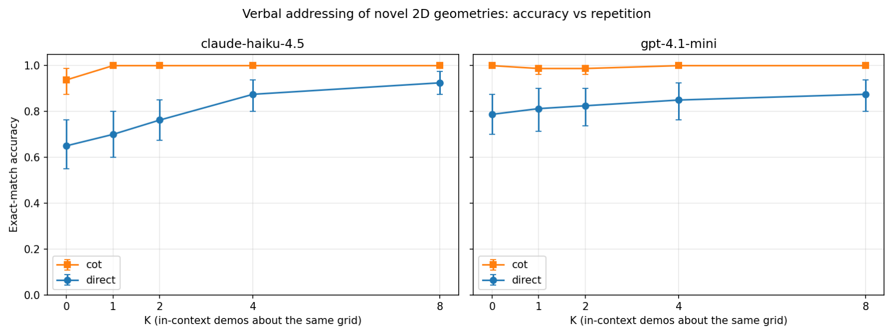

Figure 1: Accuracy on GRIDADDRESS as a function of in-context repetition K, for two models $\times$ two prompting regimes. Shaded bands are 95% bootstrap CIs (n=80). COT is at or near ceiling across all K for both models; DIRECT prompting rises with K but plateaus well below ceiling.

4.3 Effect of repetition

Table 3 contrasts K=0 and K=8 within each reasoning condition. For DIRECT prompting, repetition matters substantially (especially for CLAUDE-HAIKU-4.5, $0.650 \to 0.925$, $p < 10^{-6}$); for COT, both models are already at or near ceiling at K=0 and there is no detectable trend. Spearman ρ on K vs. correctness confirms this: CLAUDE-HAIKU-4.5 DIRECT shows ρ=0.249 (p=4.8×10^{-7}), while COT trends are weaker and uniformly toward the ceiling.

4.4 Where does DIRECT fail? Per-question-type breakdown

Table 4 pools across K and reports the error rate for each question type, and Figure 2 plots the same numbers as a heatmap. The pattern is unmistakable: DIRECT prompting handles random-access queries (POINT, LOOKUP, NEIGHBOR) at near-ceiling accuracy, and falls off a cliff on the two aggregation queries (COUNT, ROWMAX). For GPT-4.1-MINI, the DIRECT COUNT error rate is 0.675—54 errors out of 80 trials—and on inspection 50 of those 54 are off-by-one—the model registers that the token is present but loses track of how many. For CLAUDE-HAIKU-4.5, DIRECT ROWMAX errors reach 0.537. COT eliminates virtually all of these errors.

4.5 Cost-normalised view

COT uses 20–80$\times$ more output tokens than DIRECT (GPT-4.1-MINI: $2 \to 125$ tokens/call; CLAUDE-HAIKU-4.5: $8 \to 180$ tokens/call). Figure 3 plots accuracy against mean output tokens per response. Even at the most generous tokens-per-call accounting, the absolute accuracy gap (12–35 percentage points) is too large for DIRECT to be preferable on this task at realistic token prices.

4.6 MINIARC sanity check

Table 5 reports the rule-induction control. On MINIARC, GPT-4.1-MINI shows the *opposite* ordering from our addressing task (DIRECT>COT), consistent with the curse-of-CoT finding. CLAUDE-HAIKU-4.5 exhibits a prompt-compliance failure under DIRECT (it produces reasoning prose that our grid parser cannot extract from the requested single-grid format), so the CLAUDE-HAIKU-4.5 DIRECT number is a floor rather than a capability estimate; we keep the row for transparency but flag

Table 2: DIRECT vs. COT paired McNemar exact tests at every (model, K) cell (n=80 per cell). Discordant pairs are (DIRECT-only correct, COT-only correct). $\star$ marks rows significant after Bonferroni correction across the ten tests (α_{family}=0.05, α_i=0.005).

Model	K	DIRECT **acc**	COT **acc**	**Discordant**	**McNemar** p	
GPT-4.1-MINI	0	0.788	1.000	(0, 17)	1.5×10^{-5}	$\star$
GPT-4.1-MINI	1	0.812	0.988	(1, 15)	5.2×10^{-4}	$\star$
GPT-4.1-MINI	2	0.825	0.988	(0, 13)	2.4×10^{-4}	$\star$
GPT-4.1-MINI	4	0.850	1.000	(0, 12)	4.9×10^{-4}	$\star$
GPT-4.1-MINI	8	0.875	1.000	(0, 10)	2.0×10^{-3}	$\star$
CLAUDE-HAIKU-4.5	0	0.650	0.938	(2, 25)	5.6×10^{-6}	$\star$
CLAUDE-HAIKU-4.5	1	0.700	1.000	(0, 24)	1.2×10^{-7}	$\star$
CLAUDE-HAIKU-4.5	2	0.763	1.000	(0, 19)	3.8×10^{-6}	$\star$
CLAUDE-HAIKU-4.5	4	0.875	1.000	(0, 10)	2.0×10^{-3}	$\star$
CLAUDE-HAIKU-4.5	8	0.925	1.000	(0, 6)	3.1×10^{-2}	

Table 3: K=0 vs. K=8 within reasoning condition (paired McNemar). $\star$ marks Bonferroni-significant rows (α_{family}=0.05, four tests, α_i=0.0125). Repetition helps DIRECT prompting; COT is at ceiling so adding demos has no effect.

Model	Reasoning	Acc K=0	Acc K=8	Discordant	McNemar p
GPT-4.1-MINI	DIRECT	0.788	0.875	(1, 8)	3.9×10^{-2}
GPT-4.1-MINI	COT	1.000	1.000	(0, 0)	1.000
CLAUDE-HAIKU-4.5	DIRECT	0.650	0.925	(0, 22)	4.8×10^{-7} $\star$
CLAUDE-HAIKU-4.5	COT	0.938	1.000	(0, 5)	6.3×10^{-2}

it. The take-away is that our basic setup reproduces the curse-of-CoT direction in the regime where the literature predicts it, and the inverted ordering on GRIDADDRESS is therefore not an artefact of the harness.

5 Discussion

5.1 Why does COT help here when "curse of CoT" says it should not?

The mechanism is task shape, not task novelty. GRIDADDRESS is not rule induction. The grid is given verbatim and the question asks for a verifiable property of that grid; the model does not have to infer a transformation. For POINT, LOOKUP, and NEIGHBOR the model can index into the prompt directly, and DIRECT prompting succeeds at $\geq 95\%$ accuracy. For COUNT and ROWMAX the model has to scan, accumulate, and aggregate—exactly the operations that Sprague et al. [2025] identify as the regime where CoT actually buys accuracy. The fact that DIRECT COUNT errors are almost all off-by-one is the textbook failure mode that an externalised reasoning trace was designed to fix.

This refines, rather than contradicts, the curse-of-CoT story [Zheng et al., 2025]. The MINIARC control reproduces DIRECT>COT in the rule-induction regime where the curse applies; the GRIDADDRESS result shows the ordering inverts as soon as the query becomes a scan over the grid. The implication for ICL benchmarking is that pooling rule-induction and addressing queries into a single "pattern-induction" bucket conflates two very different effects.

5.2 Repetition does not bridge the aggregation gap

Table 4 reveals that even at K=8 same-grid demos, GPT-4.1-MINI DIRECT only reaches 0.44 on COUNT and CLAUDE-HAIKU-4.5 DIRECT reaches 0.875. The demonstration pool deliberately includes other COUNT examples about the same grid with correct answers, but the model still cannot generalise the counting procedure to a new token within the same grid. Repetition lifts *base accuracy* on the random-access queries, but it does not install a counting procedure—an observation consistent

Table 4: Error rate by query type, pooled across K. Random-access queries (POINT, LOOKUP, NEIGHBOR) are at near-ceiling for both models under DIRECT prompting; aggregation queries (COUNT, ROWMAX) account for almost the entire DIRECT–COT gap. Bold marks failure modes (error rate > 0.1).

Query	GPT-4.1-MINI DIRECT	GPT-4.1-MINI COT	CLAUDE-HAIKU-4.5 DIRECT	CLAUDE-HAIKU-4.5 COT
POINT	0.000	0.000	0.075	0.013
LOOKUP	0.000	0.013	0.087	0.000
NEIGHBOR	0.050	0.000	0.025	0.000
COUNT	**0.675**	0.013	**0.362**	0.000
ROWMAX	**0.125**	0.000	**0.537**	0.050

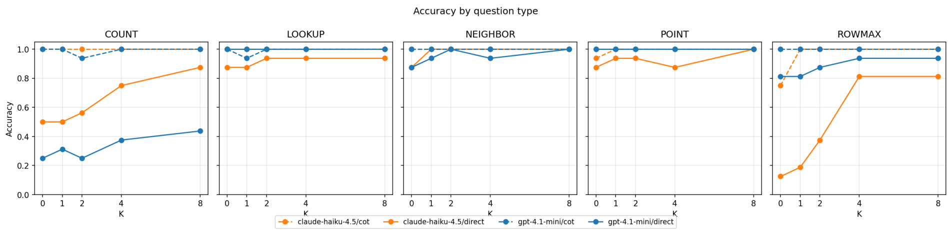

Figure 2: Per-query-type error rate for each model $\times$ reasoning condition, pooled across K. Aggregation queries (COUNT, ROWMAX) account for almost the entire DIRECT–COT gap; random-access queries are at near-ceiling under DIRECT.

with the surface self-reinforcement view of Yan et al. [2024] but at odds with the strong reading of Liu et al. [2025], which would predict that demonstrations bind a symbolic understanding of the grid that transfers across queries.

5.3 Practical takeaway

For workflows that ground LLM decisions in small text grids (UI dumps, board games, ARC-style puzzles, tabular layouts), enabling COT is a strictly cheaper reliability lever than padding the prompt with identical-grid demonstrations: it closes the aggregation gap that demonstrations cannot, and the token-cost penalty is dominated by the accuracy gain at any realistic price ratio (figure 3).

5.4 Limitations

Two models only. Both GPT-4.1-MINI and CLAUDE-HAIKU-4.5 are relatively small, "fast" production models. Larger reasoning models (o-series, Sonnet 4.5 with extended thinking, Gemini 2.5 Pro) plausibly close more of the gap even without explicit COT; the addressing gap might shrink as base capacity grows.

Single grid size. We evaluate at 7×7 only. Bai et al. [2025] report a 42–84% accuracy drop as grids grow; our results may be optimistic for larger grids, particularly for COUNT and ROWMAX where the scan length grows quadratically.

Token alphabet is colour-coded. The alphabet $\{R, B, G, Y\}$ carries colour semantics in pre-training. Truly arbitrary symbols (non-ASCII glyphs, random Unicode) may degrade either condition.

Strict exact-match scoring. We parse and score only the model's *final answer*, even when COT intermediate reasoning matches the gold. A well-reasoned response that fumbles the final-answer line counts as wrong; this likely slightly under-states COT accuracy.

Power for fine-grained contrasts. With n=80 paired observations per cell, contrasts at the ceiling (e.g., COT K=1 vs. K=2) are noisy: a single flipped item can move the estimate by 1.25 percentage points.

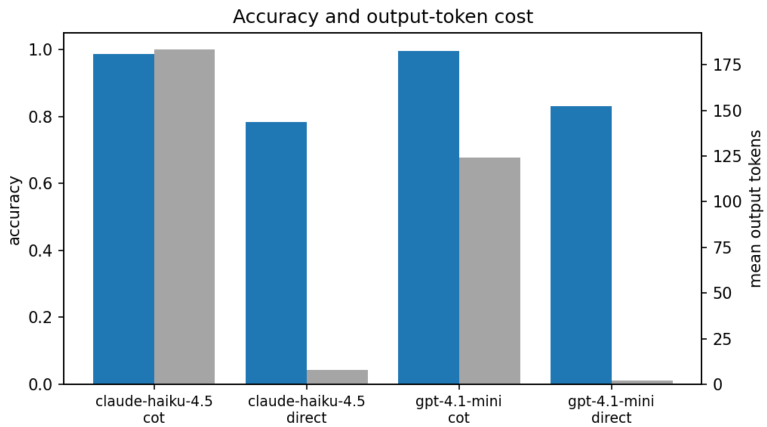

Figure 3: Accuracy on GRIDADDRESS vs. mean output tokens per response. COT uses substantially more tokens but the accuracy gain dominates at any realistic price ratio. Points show each $(\text{model}, \text{reasoning}, K)$ cell.

Table 5: Secondary MINIARC sanity check (n=25, no repetition manipulation). GPT-4.1-MINI reproduces curse-of-CoT (DIRECT>COT); CLAUDE-HAIKU-4.5 DIRECT = 0/25 reflects a prompt-compliance failure (the model emitted reasoning prose our grid parser could not extract), not a capability finding.

Model	Direct	CoT
GPT-4.1-MINI	**0.240**	0.120
CLAUDE-HAIKU-4.5	0.000†	**0.440**

†Prompt-compliance failure; see text.

MINIARC compliance. CLAUDE-HAIKU-4.5 ignored the "DIRECT only, no explanation" instruction on MINIARC and produced reasoning prose our grid parser could not extract. The MINIARC numbers should therefore be read as direction-of-effect, not as a benchmark. The GRIDADDRESS format had no such compliance issue.

No vision modality. All grids are text. The perceptual encoding story behind the forgotten-polygons line [Rudman et al., 2025] does not apply directly to our setting.

6 Conclusion

We asked whether large language models can verbally address novel 2D geometries presented as text without an explicit reasoning trace, and whether repeating the same grid in context substitutes for reasoning. On GRIDADDRESS—a controlled $2{\times}2{\times}5$ benchmark over GPT-4.1-MINI, CLAUDE-HAIKU-4.5, two prompting regimes, and five levels of in-context repetition—the answer is *partially*. Random-access queries (POINT, LOOKUP, NEIGHBOR) are addressable without reasoning at $\geq 95\%$ accuracy for both tested models, and a small amount of in-context repetition raises floor accuracy further. Aggregation queries (COUNT, ROWMAX) are *not* addressable without reasoning: DIRECT prompting fails 12–68% of the time depending on the model and query type, and eight same-grid demonstrations only partially close the gap. COT closes it almost entirely from K=0 onward, and the paired McNemar tests confirm the gap at every cell of our design.

Key takeaway. The "curse of CoT" generalisation that direct prompting dominates on in-context tasks does not extend to verbal addressing of novel geometries; the curse holds for rule induction (which our MINIARC control reproduces) and inverts for scan-and-aggregate addressing. Pooling these two regimes into a single ICL bucket misses the effect.

Future work. Several follow-ups extend the design naturally: (i) scale grids to $10{\times}10$ and $15{\times}15$ to test the Bai et al. [2025] degradation curve under both reasoning regimes; (ii) sweep the thinking-token budget on a model with explicit extended thinking (e.g., Sonnet 4.5) to see whether the COUNT/ROWMAX failure decays smoothly with thinking tokens; (iii) replace COUNT with a bounded-arithmetic variant ("how many more X than Y?") to distinguish counting failures from arithmetic failures; (iv) cross-test with a structured representation (PVD-style vertex lists or SVG [Wang et al., 2025]) as a third axis to see whether representation eliminates the DIRECT–COT gap entirely.

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